



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

**2014
ASSESSMENT TASK 1**

Mathematics Extension 2

General Instructions

- **Working time – 50 minutes**
Reading time – 5 minutes
- Write on both sides of the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators only
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**. (The multiple choice section is answered in a grid within the question booklet)
- Attempt all questions.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Lam
- ☐ Ms Ziazaris
- ☐ Mr Ireland

Student Number:

(To be used by the exam markers only.)

Question	1-5	6	7	8	Total	Percent
Mark	$\frac{\quad}{5}$	$\frac{\quad}{16}$	$\frac{\quad}{15}$	$\frac{\quad}{14}$	$\frac{\quad}{50}$	$\frac{\quad}{100}$

Section I: Objective Response

Mark your answers on the multiple choice box on the opposite page.

Marks

- 1 Suppose $z = 1 + 3i$ and $w = 2 - i$. Find $z\bar{w}$ in $x + iy$ form: 1

(A) $5 - 5i$ (B) $5 + 5i$ (C) $-1 + 7i$ (D) $5 + 7i$

- 2 What is $-\sqrt{3} + i$ expressed in modulus-argument form? 1

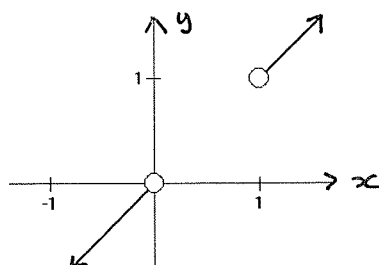
(A) $\sqrt{2}(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ (B) $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 (C) $\sqrt{2}(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$ (D) $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$

- 3 Given $(2i + 1)$ is a root of the equation $x^3 - 4x^2 + 9x - 10 = 0$ then another root is: 1

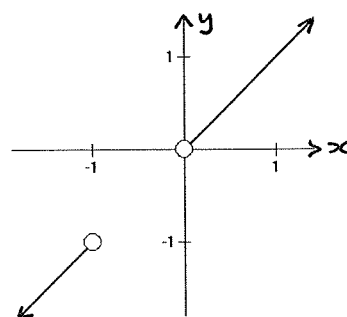
(A) 2 (B) 5 (C) $2i - 1$ (D) 10

- 4 Which of the following represents the locus $\arg z = \arg(z - [1 + i])$? 1

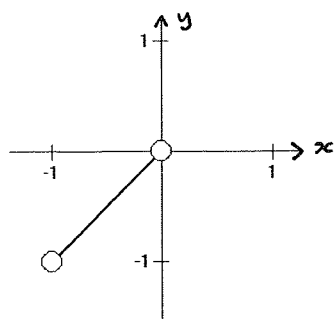
(A)



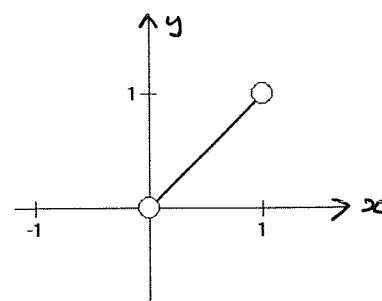
(B)



(C)

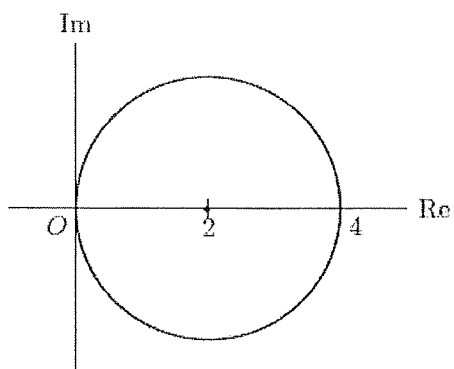


(D)



5 Which of the following is the equation of the circle shown below?

1



(A) $(z+2)(\bar{z}+2) = 4$

(B) $(z-2)(\bar{z}-2) = 4$

(C) $(z+2i)(\bar{z}-2i) = 4$

(D) $(z+2)(\bar{z}-2) = 4$

Answer grid for Section I

Mark answers to Section I by fully blackening the correct circle

1	Ⓐ	Ⓑ	Ⓒ	Ⓓ
2	Ⓐ	Ⓑ	Ⓒ	Ⓓ
3	Ⓐ	Ⓑ	Ⓒ	Ⓓ
4	Ⓐ	Ⓑ	Ⓒ	Ⓓ
5	Ⓐ	Ⓑ	Ⓒ	Ⓓ

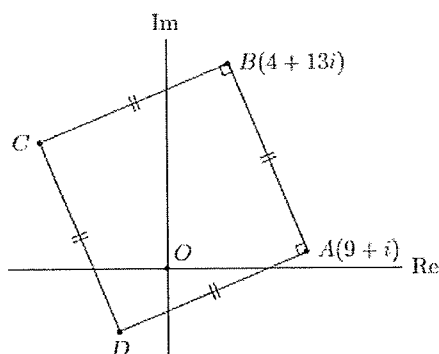
End of Section I.

Examination continues overleaf.

Section II: Short Answer

Question 6 (16 marks)	Commence a NEW page	Mark
(a) (i) Simplify $(-2i)^3$.		1
(ii) Hence show carefully on an Argand diagram all complex numbers z such that $z^3 = 8i$.		2
(iii) Express the three cube roots of $8i$ in the form $x + iy$.		2
(b) Let ω be one of the non-real cube roots of 1.		
(i) Show that $1 + \omega + \omega^2 = 0$		1
(ii) Evaluate $(-1 + \omega + \omega^2)(1 - \omega + \omega^2)(1 + \omega - \omega^2)$		2
(iii) Evaluate $(1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11})$		2
(c) Simplify $\frac{\cos 3\theta + i \sin 3\theta}{\cos 2\theta - i \sin 2\theta}$		2
(d) (i) On a single Argand diagram, sketch the locus where $ z - 3i \leq 2$ and $\arg(z + 1) \leq \frac{\pi}{4}$ apply.		2
(ii) Find the value of $\arg z$ where $\arg z$ is a minimum. (You may leave answer in exact form)		2

- (a) In the following diagram, $ABCD$ is a square in the complex plane. Vertices A and B represent the complex numbers $9+i$ and $4+13i$ respectively.



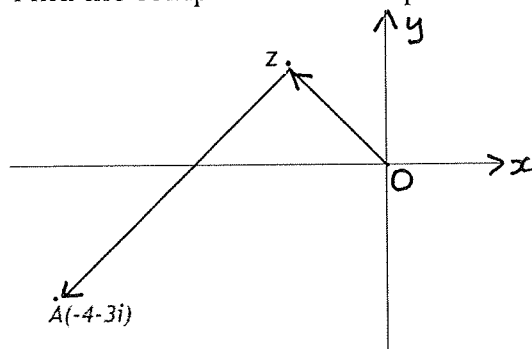
Find the complex numbers represented by:

- (i) The vector $\overrightarrow{AB}$ 1
 (ii) The vertex D 2

- (b) It can be shown $\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$ and $\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$.

- (i) Deduce that $\tan 3\theta = \frac{\tan^3 \theta - 3\tan \theta}{3\tan^2 \theta - 1}$ 1
 (ii) Hence find the roots of $x^3 - 3x^2 - 3x + 1 = 0$ 3
 (iii) Without using a calculator, evaluate $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12}$. 2

- (c) The point A represents the complex number $-4-3i$. $\angle OZA = 90^\circ$ and $ZA = 2 \times OZ$. Find the complex number represented by the point Z . 3



- (d) Sketch the locus of z if $\frac{z-1}{z-2i}$ is purely imaginary. 3

(a) It is given that $5 + 6i$ is a zero of $P(x) = 2x^3 - 19x^2 + 112x + d$, where d is real.

(i) What are the other two zeroes of $P(x)$? 2

(ii) Find the value of d . 2

(b) When $P(x) = x^4 + ax^2 + bx$ is divided by $x^2 + 1$ the remainder is $x + 2$.

Find the values of a and b . 2

(c)

(i) Suppose that the polynomial $P(x)$ has a double root at $x = \alpha$.

Prove that $P'(x)$ also has a root at $x = \alpha$. 2

(ii) The polynomial $P(x) = x^4 + ax^2 + bx + 36$ has a double root at $x = 2$.

Find the values of a and b . 2

(d) If the equation $x^3 - 3x^2 - x + 2 = 0$ has roots α, β, γ then find the

polynomial equation with roots $2\alpha + \beta + \gamma$, $\alpha + 2\beta + \gamma$, and $\alpha + \beta + 2\gamma$. 3

(e) How many positive integers n are there such that $n + 3$ divides $n^2 + 7$ without a remainder? 1

END OF EXAM

Q1 - C

Q2 - D

Q3 - A

Q4 - A

Q5 - B

[Reasoning for Section 1:

$$(1) \quad \begin{aligned} z\bar{w} &= (1+3i)(2+i) \\ &= 2+i+6i-3 \\ &= -1+7i \quad \therefore (C) \end{aligned}$$

$$(2) \quad \begin{aligned} -\sqrt{3} + i &= 2\left(-\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2}\right) \\ &= 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) \quad \therefore (D) \end{aligned}$$

$$(3) \quad \begin{aligned} \text{Let roots be } 1+2i, 1-2i, \alpha \\ \text{Sum of roots is } 4 \quad \therefore 2+\alpha=4 \\ \quad \quad \quad \therefore \alpha=2 \quad \therefore (A) \end{aligned}$$

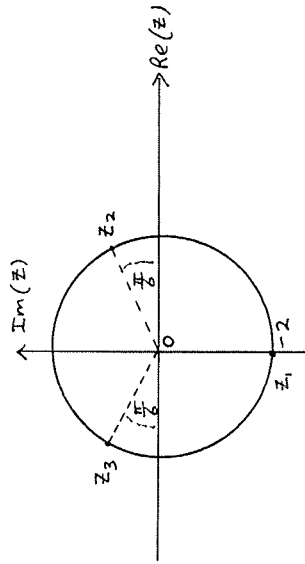
$$(4) \quad \begin{aligned} \arg(z-z_1) &= \arg(z-z_2) \text{ is the} \\ \text{locus of all points dividing } z_1 \text{ \& } z_2 \\ \text{externally.} \\ \therefore \arg(z-0) &= \arg(z-(1+i)) \text{ is (A)} \end{aligned}$$

$$(5) \quad \begin{aligned} \text{Circle is } (x-2)^2 + y^2 &= 4 \\ \text{i.e. } (x^2+y^2) - 2(2x) + 4 &= 4 \\ \text{i.e. } (z-2)(\bar{z}-2) &= 4 \quad \therefore (B) \end{aligned}$$

[Q6]

$$(a) \quad (i) \quad (-2i)^3 = -8i^3 = 8i$$

(ii)



[One solution, by (i), is at $-2i$, and the other two lie on the circle radius 2, centre (0,0), $\frac{2\pi}{3}$ radians apart.]

$$(iii) \quad \begin{aligned} \text{From (ii), } z_1 &= 0-2i \\ z_2 &= 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) \\ \therefore z_2 &= \sqrt{3} + i \\ z_3 &= 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) \\ \therefore z_3 &= -\sqrt{3} + i \end{aligned}$$

(ALT: candidates could use De Moivre $\rightarrow z = 2\left[\cos\left(\frac{2n\pi}{3} + \frac{\pi}{6}\right)\right]$, etc.
or Sum of 2 cubes $\rightarrow z^3 + 8i^3 = 0$, etc.)

Q6 continued(b) (i) we have $\omega^3 = 1$

$$\therefore \omega^3 - 1 = 0$$

$$\therefore (\omega - 1)(\omega^2 + \omega + 1) = 0$$

since ω is non-real, $\therefore \omega^2 + \omega + 1 = 0$.

$$(ii) (-1 + \omega + \omega^2)(1 - \omega + \omega^2)(1 + \omega - \omega^2)$$

$$= (-2)(-2\omega)(-2\omega^2)$$

$$= -8\omega^3$$

$$= -8$$

$$(iii) (1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11})$$

$$= (1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4)(1 + \omega^5)$$

$$= (-\omega^2)(-\omega)(-\omega)(-\omega)(-\omega)$$

$$= -\omega^6$$

$$= -1$$

$$(c) \frac{\cos 3\theta + i \sin 3\theta}{\cos 2\theta - i \sin 2\theta} = \frac{\cos 3\theta + i \sin 3\theta}{\cos(-2\theta) + i \sin(-2\theta)}$$

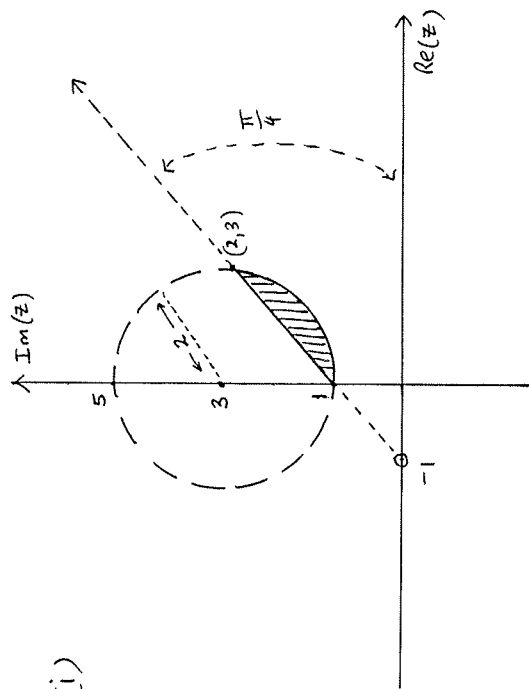
$$= \frac{(\cos \theta + i \sin \theta)^3}{(\cos \theta + i \sin \theta)^{-2}}$$

$$= (\cos \theta + i \sin \theta)^5$$

$$= \cos 5\theta + i \sin 5\theta.$$

Q6 continued

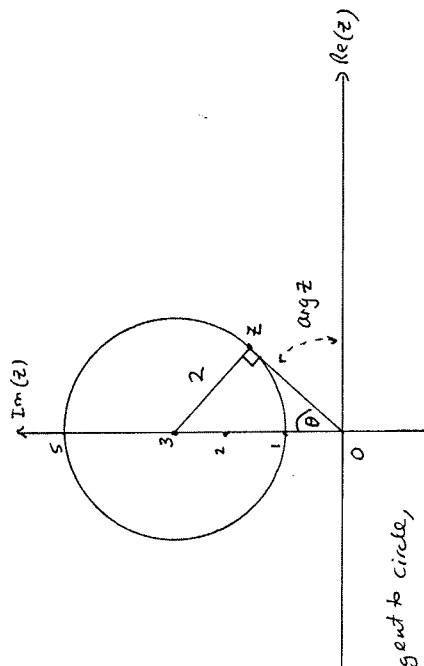
(d) (i)



✓ correctly displays one locus

✓ correctly shows both loci with shading

(ii)



OZ is tangent to circle,

$$\therefore \sin \theta = \frac{2}{3}$$

$$\therefore \arg z = \frac{\pi}{2} - \sin^{-1}\left(\frac{2}{3}\right)$$

$$(\doteq 0.841069).$$

✓ progress towards solution

✓ correct answer

Q7

(a) (i) $\vec{AB}$ represents $-5 + 12i$ (ii) $\vec{AD}$ is $\vec{AB}$ rotated 90° counter-clockwise. $\therefore \vec{AD}$ represents $i(-5 + 12i)$

$$= -12 - 5i$$

 $\therefore$ vertex D represents $-3 - 4i$.

$$\begin{aligned}
 \text{(b) (i) } \tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\
 &= \frac{3\cos^2\theta \sin\theta - \sin^3\theta}{\cos^3\theta - 3\cos\theta \sin^2\theta}
 \end{aligned}$$

$$= \frac{3 \tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} \quad \left(\begin{array}{l} \text{Dividing} \\ \text{by} \\ \cos^3\theta \end{array} \right)$$

$$= \frac{\tan^3\theta - 3\tan\theta}{3\tan^2\theta - 1}, \text{ as required.}$$

(ii) Let $x = \tan\theta$.

The equation becomes:

$$\tan^3\theta - 3\tan^2\theta - 3\tan\theta + 1 = 0$$

$$\therefore \tan^3\theta - 3\tan^2\theta = 3\tan^2\theta - 1$$

$$\therefore \frac{\tan^3\theta - 3\tan\theta}{3\tan^2\theta - 1} = 1$$

$$\therefore \tan 3\theta = 1 \quad (\text{from (i)})$$

$$3\theta = \frac{\pi}{4} + n\pi \quad (n \text{ an integer})$$

$$\theta = \frac{4n+1 \cdot \pi}{12}$$

✓ correct answer

✓ vector $\vec{AD}$

✓ correct answer

✓ shows all working

✓ Correctly applies part (i)

✓ correct θ 's.

Q7 – continued

(b) (ii) (continued)

$$n=0 \rightarrow \theta = \frac{\pi}{12}$$

$$n=1 \rightarrow \theta = \frac{5\pi}{12}$$

$$n=2 \rightarrow \theta = \frac{9\pi}{12} = \frac{3\pi}{4}$$

$$\text{So } x = \tan \frac{\pi}{12}, \tan \frac{5\pi}{12}, \tan \frac{3\pi}{4} = -1.$$

(iii) by sum of roots,

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} - 1 = 3$$

$$\therefore \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$$

$$\text{(c) } \vec{OZ} + \vec{ZA} = \vec{OA}$$

$$\text{Let } z = x + iy$$

$$\therefore (x + iy) + 2i(x + iy) = -4 - 3i$$

$$\therefore x - 2y + i(2x + y) = -4 - 3i$$

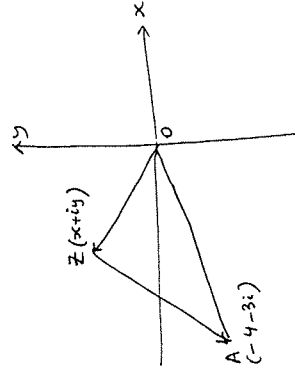
 $\therefore$ Equating real & imaginary parts,

$$\begin{cases} x - 2y = -4 \\ 2x + y = -3 \end{cases}$$

Solving simultaneously,

$$\begin{aligned} x &= -2 \\ y &= 1 \end{aligned}$$

$$\therefore Z = -2 + i.$$

[see over for alternate approaches $\rightarrow$][Note: algebraic solutions giving $-1, 2 \pm 3$ receive 2 marks].

✓ Gives 3 correct roots

✓ attempts to use sum of roots

✓ correct answer

✓ Correct start

✓ Correct working

✓ correct answer

[ALTERNATIVE SOLUTIONS to 7(c):

[1] $\vec{OZ} + \vec{ZA} = \vec{OA}$

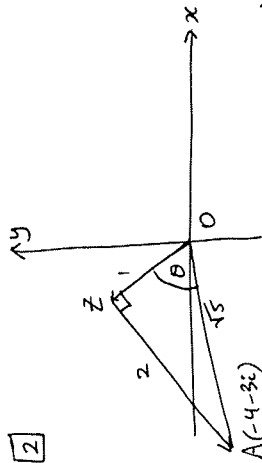
But $\vec{ZA}$ is $\vec{OZ}$ rotated $\frac{\pi}{2}$ counterclockwise and stretched by a factor of 2.

$\therefore z + 2i \cdot z = -4 - 3i$

$z(1 + 2i) = -4 - 3i$

$z = \frac{-4 - 3i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i}$

$= \frac{-4 + 8i - 3i - 6}{5} \quad \therefore z = -2 + i.$



$$\begin{aligned} \vec{OA} &= \sqrt{5} \cdot \text{cis } \theta \cdot \vec{OZ} \\ &= \sqrt{5} \text{cis}(\tan^{-1} 2) \cdot \vec{OZ} \\ &= \sqrt{5} \left[\frac{1}{\sqrt{5}} + i \frac{2}{\sqrt{5}} \right] \cdot \vec{OZ} \\ &= (1 + 2i) \cdot \vec{OZ} \end{aligned}$$

$\therefore z = \frac{-4 - 3i}{1 + 2i} = -2 + i$

[3] $\vec{OZ} = \frac{1}{\sqrt{5}} \cdot \text{cis}(-\theta) \cdot \vec{OA}$

$\curvearrowright$ clockwise rotation here

$= \frac{1}{\sqrt{5}} \left(\frac{1}{\sqrt{5}} - i \frac{2}{\sqrt{5}} \right) \cdot (-4 - 3i)$

$= \left(\frac{1}{5} - i \frac{2}{5} \right) (-4 - 3i)$

$= -2 + i. \#$

]

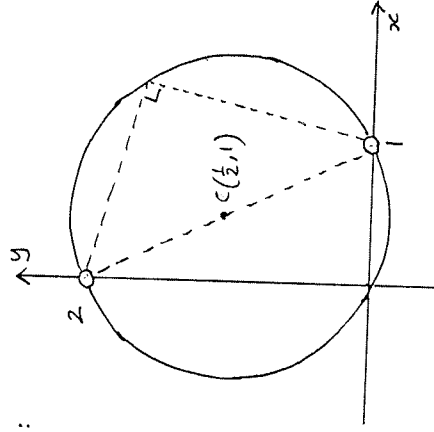
[Q7] continued

(d) $\frac{z-1}{z-2i}$ is purely imaginary.

$\therefore \arg\left(\frac{z-1}{z-2i}\right) = \pm \frac{\pi}{2}.$

This is a circle on diameter AB, where $A = (1, 0)$ and $B = (0, 2)$, excluding the points A and B.

Hence:



Needs:

- Corred circle locus, through $(0, 2)$, $(1, 0)$ and $(0, 0)$.
- open circle at $(0, 2)$
- centre shown

[ALT: Let $z = x + iy$

$\therefore \frac{z-1}{z-2i} = \frac{(x-1)+iy}{x+i(y-2)} \times \frac{x-i(y-2)}{x-i(y-2)} \quad (z \neq 2i)$

$= \frac{x(x-1) + y(y-2) + i[xy - (x-1)(y-2)]}{x^2 + (y-2)^2}$

Now, Re part = 0, $\therefore x(x-1) + y(y-2) = 0$

$\therefore (x - \frac{1}{2})^2 + (y-1)^2 = \frac{5}{4}$

which is a circle, centre $(\frac{1}{2}, 1)$, radius $\frac{\sqrt{5}}{2}$,

excluding $(0, 2)$, as shown above.

[Note: texts differ on whether $(0, 2)$ is regarded as 'purely imaginary', as it lies on both Re and Im axes. See Patel p.114, Lee p.52. Hence difference in the two approaches at $(1, 0)$, & openably either way.

✓ first na here.

✓ complete square to get circle

✓ correct sketch (as above)

Q8

(a) $P(x) = 2x^3 - 19x^2 + 112x + d$, (d real).

(i) Since coefficients real,
another zero is $5 - 6i$.

Let 3rd coefficient be α .

Sum of coefficients $\Rightarrow (5+6i) + (5-6i) + \alpha = \frac{19}{2}$

$\therefore \alpha = -\frac{1}{2}$

Thus other zeroes are $5 - 6i$, $-\frac{1}{2}$.

(ii) product of zeroes $= -\frac{d}{2}$

$\therefore (5+6i)(5-6i)(-\frac{1}{2}) = -\frac{d}{2}$

$\therefore d = 5^2 + 6^2 = 61$

$\therefore d = 61$

(b) $P(x) = x^4 + ax^2 + bx = (x^2 + 1) \cdot Q(x) + x + 2$

Sub. in $x = i \Rightarrow$

$\therefore i^4 + a \cdot i^2 + bi = 0 + i + 2$

$\therefore (1-a) + bi = 2 + i$

Equate real & imaginary parts,

$1-a = 2$
 $b = 1$

$\therefore a = -1$
 $b = 1$

Q8 continued

(c)

(i) If $P(x)$ has double root at $x = \alpha$,

then $P(x) = (x-\alpha)^2 \cdot Q(x)$, where $Q(x) \neq 0$

$\therefore P'(x) = Q(x) \cdot 2(x-\alpha) + (x-\alpha)^2 \cdot Q'(x)$

$= (x-\alpha) [2 \cdot Q(x) + (x-\alpha) \cdot Q'(x)]$

So $P'(\alpha) = 0 [2 \cdot Q(\alpha) + 0 \cdot Q'(\alpha)]$

$= 0$

$\therefore x = \alpha$ is also a root of $P'(x)$.

(ii) $P(x) = x^4 + ax^2 + bx + 36$

$\therefore P'(x) = 4x^3 + 2ax + b$

$P(2) = P'(2) = 0$ as $x=2$ is a double root of $P(x)$.

$\therefore 16 + 4a + 2b + 36 = 0$

$\therefore 2a + b = -26 \quad \text{--- (1)}$

and $32 + 4a + b = 0$

$\therefore 4a + b = -32 \quad \text{--- (2)}$

(2) - (1) $\Rightarrow 2a = -6$

$\therefore a = -3$

$b = -20$

Q8 – Continued

$$(d) \quad \alpha + \beta + \gamma = 3$$

$\therefore$ we want roots $\alpha+3, \beta+3, \gamma+3$.

$\therefore$ equation is

$$(x-3)^3 - 3(x-3)^2 - (x-3) + 2 = 0$$

$$\therefore x^3 - 9x^2 + 27x - 27 - 3x^2 + 18x - 27 - x + 3 + 2 = 0$$

$$\therefore x^3 - 12x^2 + 44x - 49 = 0$$

$$(e) \quad \begin{array}{r} n-3 \\ n+3 \overline{) n^2 + 0n + 7} \\ \underline{n^2 + 3n} \\ -3n + 7 \\ \underline{-3n - 9} \\ 16 \end{array}$$

$$\therefore n^2 + 7 = (n+3)(n-3) + 16$$

So, for $n+3$ to divide n^2+7 with no remainder, we must have $n+3$ divides 16.

Since $n > 0$, $\therefore n+3 = 4, 8, \text{ or } 16$

$$n = 1, 5, \text{ or } 13$$

$\therefore 3$ integers.